

Week 2 Report — Vortex Tech Cyber Security Internship

Introduction

Week 2 of the Cyber Security Internship focused on two complementary skill areas: building a piece of defensive security tooling from scratch, and performing hands-on network reconnaissance using industry-standard tools. The objective was to move beyond theory and produce working, testable artifacts — a Python program that actively helps users choose safer passwords, and a documented Nmap assessment of real (authorized) network targets — while practicing the habits of a working security professional: authorization awareness, evidence-based reporting, and clear remediation guidance.

This document serves as the central navigation page for the Week 2 submission. Full implementation details, algorithms, and raw findings are intentionally kept in their respective component reports rather than repeated here.

Assignment Summary

Password Strength Checker A command-line Python tool was built to evaluate passwords against length requirements, character-type variety (uppercase, lowercase, digits, special symbols), and a blacklist of common/leaked passwords. Rather than a simple pass/fail result, the tool produces a five-tier strength rating with a transparent score and specific, personalized feedback explaining what to improve.

Nmap Port Scan Two authorized network scans were performed: a

service and OS-detection scan against a personal localhost machine, and a network sweep across a personal home network. The exercise involved identifying live hosts, open and closed ports, associated services, and documenting the security implications of what was found — along with practical recommendations to reduce exposure.

Project Components

- [password_checker_report.md](#) — full design, algorithm, and discussion of the password strength checker
- [nmap_report.md](#) — complete scan commands, output, findings, and recommendations from the network reconnaissance exercise
- [sample_outputs.md](#) — real, reproducible terminal output from 8 test passwords covering all strength tiers

Screenshots

Password Strength Checker

[Password Tests](#)

Nmap Scans

[Localhost Scan](#) [Network Scan](#)

Key Learning Outcomes

- Translating password security theory (entropy, complexity, blacklisting) into working, modular code
- Recognizing that character-complexity rules alone do not guarantee a strong password without blacklist checks
- Designing feedback systems that guide users toward better security decisions rather than simply rejecting weak input
- Practicing authorized, scoped network reconnaissance using

Nmap

- Mapping open ports to real services and reasoning about the risk each exposed service introduces
- Distinguishing between necessary and unnecessary network exposure as a core attack-surface-reduction principle
- Writing security findings and recommendations in a clear, professional, evidence-based format

Reflection

Working on these two exercises side by side made the connection between password security and network security more concrete than either topic would have been on its own. A password checker is, at its core, about reducing the odds that a single compromised credential leads to account takeover. A network scan is about understanding — from an attacker's perspective — which doors into a system are actually open, and whether they need to be. Both exercises are really asking the same underlying question: *what is exposed, and does it need to be?*

The Nmap scan reinforced this directly. Most devices on the home network had no open ports at all, while the router exposed only SSH and HTTPS — services that were actually needed for management. That is attack surface reduction in practice: not "lock everything down blindly," but "only expose what serves a real purpose, and secure whatever you do expose." The password checker mirrors that same philosophy on the credential side — a strong password reduces the odds that an otherwise well-secured system gets bypassed simply because a login was easy to guess.

This week also reinforced the importance of secure system administration as an ongoing discipline rather than a one-time setup task. Services get added and forgotten, default credentials get left in place, and firmware goes unpatched — not usually out of carelessness, but because there's no built-in prompt reminding administrators to check. Both the password checker's feedback loop and the practice of periodic Nmap scanning serve that same role: creating a repeatable habit of checking assumptions instead of trusting that "it was secure when we set it up" still holds true.

Conclusion

Week 2 combined defensive tool-building with hands-on reconnaissance to reinforce a core security principle: minimizing and securing every point of exposure, whether that exposure is a weak password or an unnecessarily open network port. The password strength checker demonstrates how thoughtful, rule-based logic can meaningfully guide users toward safer credentials, while the Nmap assessment demonstrated how straightforward, authorized reconnaissance can reveal exactly what a network exposes to the outside world. Together, these exercises provided a practical foundation in both secure software design and network security assessment — two skill sets that this internship continues to build toward a more complete, applied understanding of cybersecurity.